

UNIVERSITY OF TECHNOLOGY

Professor Course Structure

Semester VII - Data Structures & Algorithms (DSA)

Course Code: CS701

Instructor: DSA Professor

Credits: 4 (Theory + Lab)

Unit	Topic	Details	Hours
1	Advanced Trees	AVL, Red-Black Trees, Heaps	8
2	Graph Algorithms	BFS, DFS, Dijkstra, Floyd-Warshall	10
3	Greedy Algorithms	Huffman Coding, Activity Selection	6
4	Dynamic Programming	Knapsack, LCS, Matrix Chain	10
5	String Algorithms	KMP, Rabin-Karp, Trie	6
6	Advanced Topics	Amortized Analysis, NP-Completeness	6

Lab Work:

Implementation of DSA problems using C++/Java/Python.
Mini project based on real-world problem solving.

Evaluation:

Mid-Term: 30 Marks

End-Term: 50 Marks

Assignments + Lab: 20 Marks